

To: Technical Manager
From: Brandon Acosta
RE: Job Fit Summary LLM Model

Summary

I developed a Job Fit Analyzer using an LLM model. It is a retrieval-augmented generation pipeline for comparing a candidate resume against multiple job descriptions. The application has three types of output for each job description: a Skill Gap Report, Keyword Alignment, and Fit Summary.

What I Built

For each job description, the pipeline retrieves relevant context from the job description and resume, then generates one of three analysis types:

1. Skill Gap Report - Identifies matching skills, missing skills, and recommended actions to improve skill alignment.
2. Keyword Alignment - Creates a table of job keywords or phrases, marks whether the resume matches them, and provides a final match rate percentage.
3. Fit Summary - Produces a 3-4 sentence written summary explaining how well the candidate fits the role, with evidence of strengths and weaknesses.

Model Recommendation

I recommend deploying the Fit Summary output as the primary model output. The Fit Summary performed the most consistently across all three job descriptions. In the evaluation, the Fit Summary has the ability to ensure proper separation of hard/technical skills, soft skills, and preferred industry experience qualifications. The Skill Gap Report can be added as an augmentation of the Fit Summary since it provides more specific and useful recommendations when there are missing skills from the list of qualifications in the desired job description.

Business Implications

The main business value of this system is improved speed of job application, with more consistent results and tailored resume creation for job seekers. This tool can be used by students or job seekers to:

- Tailor resume before applying to jobs
- Career coaching and advising based on skill expansion recommendations
- Comparison of one resume across multiple job descriptions (to tailor job search to industry, job title, experience level, etc.)

Key Limitation

This model is highly dependent on the quality of input text for analysis. Job description inputs should be consistent, and careful to exclude unnecessary information like legal notices, and recruitment timelines. Another limitation is that the model can be too literal when evaluating soft skills. In some outputs, it treated soft skills like communication, organization, attention to detail, etc. as missing because they were not stated word-for-word in the resume.